

Hawaii SNAP Analysis: Interactive Homework

Code ▼

Using Census Data & Machine Learning to Understand Food Security

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Introduction: Food Security in Hawai'i

Food security is a critical health and equity issue. The Supplemental Nutrition Assistance Program (SNAP) helps vulnerable households afford nutritious food. But which communities have the highest rates of food assistance participation? And what factors are most strongly associated with SNAP participation?

In this assignment, you'll:

1. **Access Census data** using `tidycensus`
2. **Explore patterns** of SNAP participation across Hawai'i census tracts
3. **Build a predictive model** using Random Forests
4. **Interpret results** using SHAP (Explainable AI) techniques

Homework Requirements

- ✓ Complete all **CHALLENGE** sections
- ✓ Provide code AND written explanations
- ✓ Show your thinking - comment your code!
- ✓ Answer reflection questions at the end
- ★ Bonus: Try the extension questions for extra credit

Due Date: [Before Class, 2:30 PM pm 10/23/25]



Load Libraries

```
library(ggplot2)
library(tidycensus)
library(dplyr)
library(tidyr)
library(stringr)
library(maggl)
library(viridis)
library(randomForest)
library(treeshap)
library(mapview)
library(sf)
```



Part 1: Collect Census Data

Set Up Your Census Variables

The code below defines the Census Bureau variables we'll use. These come from the American Community Survey (ACS) 5-Year Estimates.

```
# Define ACS variables for Hawaii
# (Population, poverty, insurance, employment, housing, broadband, education, income, SNAP)

hi_vars <- c(
  # Person-level
  poverty_n          = "B17001_002",
  uninsured_u65_n    = "B27010_017",
  unemployed_n       = "B23025_005",
  pop_total          = "B01003_001",

  # Households + housing
  households_total    = "B11001_001",
  broadband_hh_n      = "B28002_004",
  renter_hh_n         = "B25003_003",
  novehicle_hh_n      = "B25044_003",
  snap_hh_n           = "B22001_002",      # <-- SNAP households

  # Education
  edu_25plus_total    = "B15003_001",
  bachelors_n         = "B15003_022",

  # Labor force + workers
  labor_force_total   = "B23025_003",
  workers_total       = "B08301_001",

  # Income level
  median_income       = "B19013_001"
)
```

Understanding Census Variables

Each variable code (like "B17001_002") refers to a specific data table and line number from the Census Bureau. The naming convention is: - **B** = Detailed table - **17001** = Table ID (in this case, poverty data) - **002** = Specific line number (usually the numerator for a percentage calculation)

For more info, check the [Census Bureau Data API documentation](#)

Pull Data from the Census API

```
# Pull core variables (long format, then pivot to wide)
hi_acs_long <- get_acs(
  geography = "tract",
```

```

state = "HI",
year = 2023,
geometry = TRUE,
variables = hi_vars
)|>
dplyr::filter(GEOID != "15003981200")

# Convert to wide format (one row per census tract)
hi_acs <- hi_acs_long %>%
  select(GEOID, variable, estimate) %>%
  pivot_wider(names_from = variable, values_from = estimate)

```

Note

Census data loaded successfully! (Output suppressed to keep document clean)

Calculate Commute Times (≥ 30 minutes)

```

# Get commute time data using table lookup
commute_tbl <- get_acs(
  geography = "tract",
  state = "HI",
  year = 2023,
  table = "B08303",
  geometry = FALSE
)|>
dplyr::filter(GEOID != "15003981200")

# Extract commute  $\geq 30$  min variable
commute_30plus <- commute_tbl %>%
  filter(variable == "B08303_004") %>%
  group_by(GEOID) %>%
  summarize(commute_30plus_n = sum(estimate, na.rm = TRUE), .groups = "drop")

```



Part 2: Calculate Percentages & Clean Data

The code below joins all the data and calculates percentages. Study how we normalize each variable!

```

# Join commute data and calculate all percentages
hi_acs <- hi_acs %>%
  left_join(commute_30plus, by = "GEOID") %>%
  mutate(
    # Person-based percentages
    poverty_rate_pct      = 100 * poverty_n      / pmax(pop_total, 1),
    uninsured_u65_pct     = 100 * uninsured_u65_n / pmax(pop_total, 1),

    # Labor-force based

```

```

unemployment_rate_pct = 100 * unemployed_n / pmax(labor_force_total, 1),

# Household-based percentages
broadband_pct = 100 * broadband_hh_n / pmax(households_total, 1),
renter_pct = 100 * renter_hh_n / pmax(households_total, 1),
no_vehicle_pct = 100 * novehicle_hh_n / pmax(households_total, 1),
snap_pct = 100 * snap_hh_n / pmax(households_total, 1),

# Education (25+)
bachelors_or_higher_pct = 100 * bachelors_n / pmax(edu_25plus_total, 1),

# Commute-based
commute_30plus_pct = 100 * commute_30plus_n / pmax(workers_total, 1)
)

```

Why `pmax(denominator, 1)` ?

This prevents division by zero errors. If a census tract has 0 people, we use 1 as the denominator instead. The `pmax()` function takes the parallel maximum - comparing each value to 1 element-wise.

CHALLENGE 1: Map SNAP Participation

Your Task

Create a map showing SNAP participation rates by census tract across Hawai'i. You should:

1. **Use** `ggplot2` with `geom_sf()` to map the data
2. **Color by** `snap_pct` (use a viridis color scale)
3. **Add labels** including title, subtitle, and data source
4. **Make it look professional** with clean theme/formatting

```

library(ggplot2)
library(viridis)
library(sf)

ggplot(data = hi_acs) +
  geom_sf(aes(fill = snap_pct), color = "white", size = 0.1) +
  scale_fill_viridis(
    name = "SNAP Participation (%)",
    option = "C",
    direction = -1,
    na.value = "gray90"
  ) +
  labs(
    title = "SNAP Participation Across Hawai'i by Census Tract",
    subtitle = "Share of households receiving Supplemental Nutrition Assistance Program (SNAP) benefits",
    caption = "Source: U.S. Census Bureau, 2023 ACS 5-Year Estimates"
  ) +

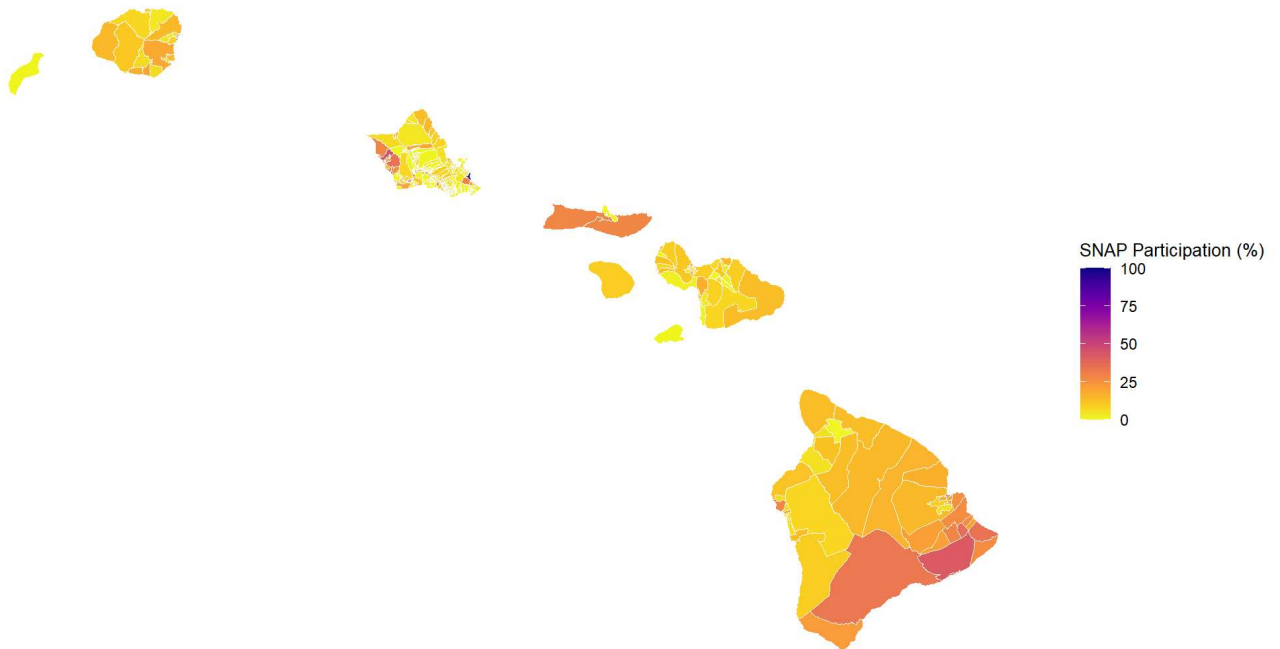
```



```
theme_minimal(base_family = "Helvetica") +
theme(
  plot.title = element_text(size = 16, face = "bold"),
  plot.subtitle = element_text(size = 12, color = "gray30"),
  plot.caption = element_text(size = 9, color = "gray40", hjust = 0),
  legend.position = "right",
  legend.title = element_text(size = 10),
  legend.text = element_text(size = 9),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  axis.text = element_blank(),
  axis.title = element_blank(),
  axis.ticks = element_blank()
)
```

SNAP Participation Across Hawai'i by Census Tract

Share of households receiving Supplemental Nutrition Assistance Program (SNAP) benefits, 2023



Source: U.S. Census Bureau, 2023 ACS 5-Year Estimates

Reflection on Your Map

CHALLENGE 2: Interactive Map with mapview

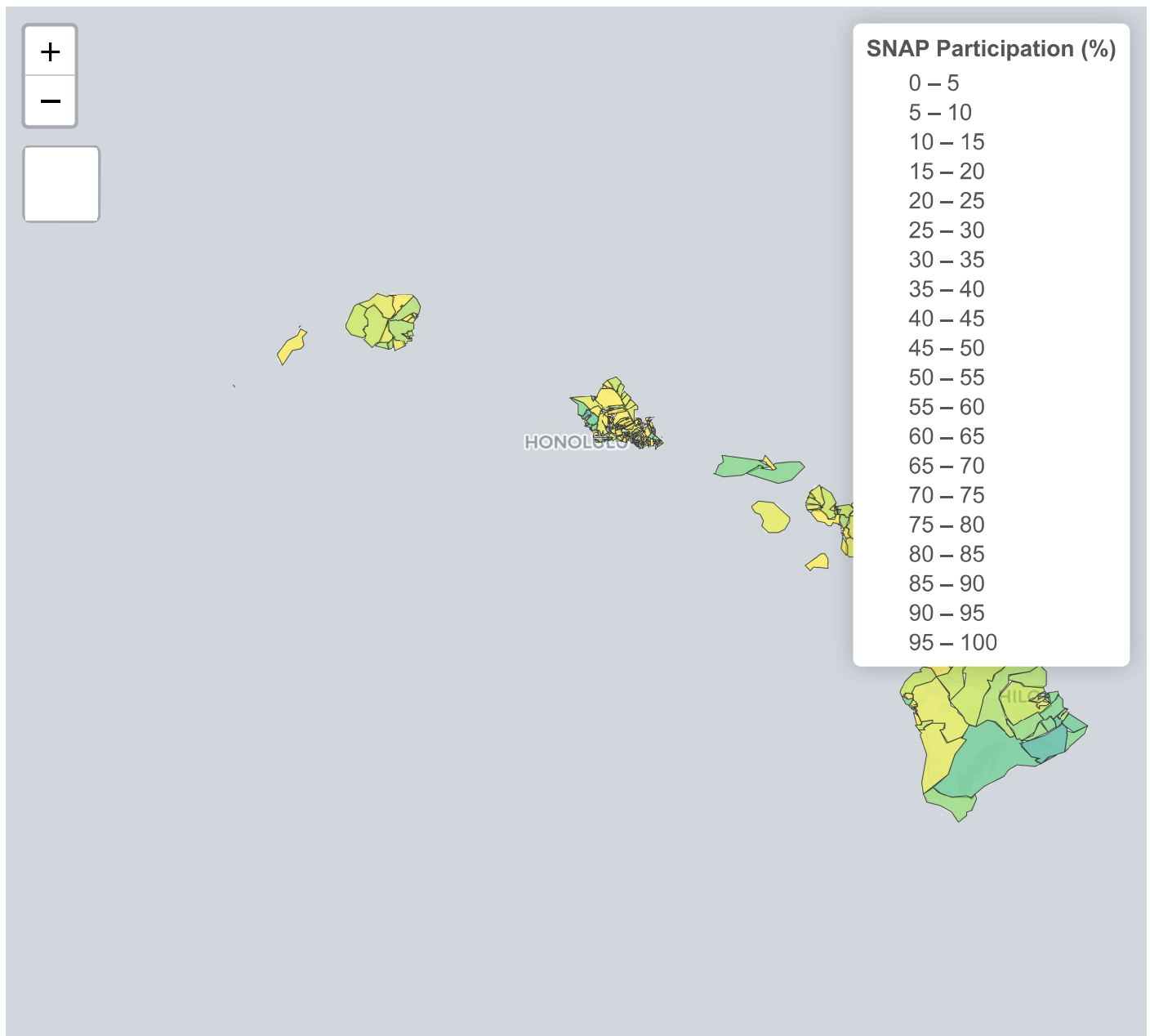
Your Task

Create an **interactive leaflet-style map** using the `mapview` package.

Hints & Resources: - The `mapview()` function creates interactive maps - Use `zcol = "snap_pct"` to color by SNAP participation - Set `legend = TRUE` to show a legend - Use `layer.name` to label your layer - The `at` parameter controls color breaks (try: `seq(0, max(...), by = 5)`) - Use `col.regions = viridis::viridis(10, direction = -1)` for colors

```
library(mapview)
library(viridis)

mapview(
  hi_acs,
  zcol = "snap_pct",           # Color by SNAP percentage
  legend = TRUE,              # Show legend
  layer.name = "SNAP Participation (%)", # Legend title
  at = seq(0, max(hi_acs$snap_pct, na.rm = TRUE), 5), # Color breaks every 5%
  col.regions = viridis(10, direction = -1), # Viridis color palette
  map.types = "CartoDB.Positron" # Clean basemap
)
```



100 km

50 mi

SNAP_Participation_____

Interactive Map Features

With `mapview`, you can: - Zoom and pan - Click on tracts to see exact values - Toggle layers on/off - Switch between map types - Export the map

Try zooming into a specific island and exploring the variation!



CHALLENGE 3: Histogram of SNAP Participation

Your Task

Create a histogram showing the distribution of `snap_pct` across all Hawaii census tracts.

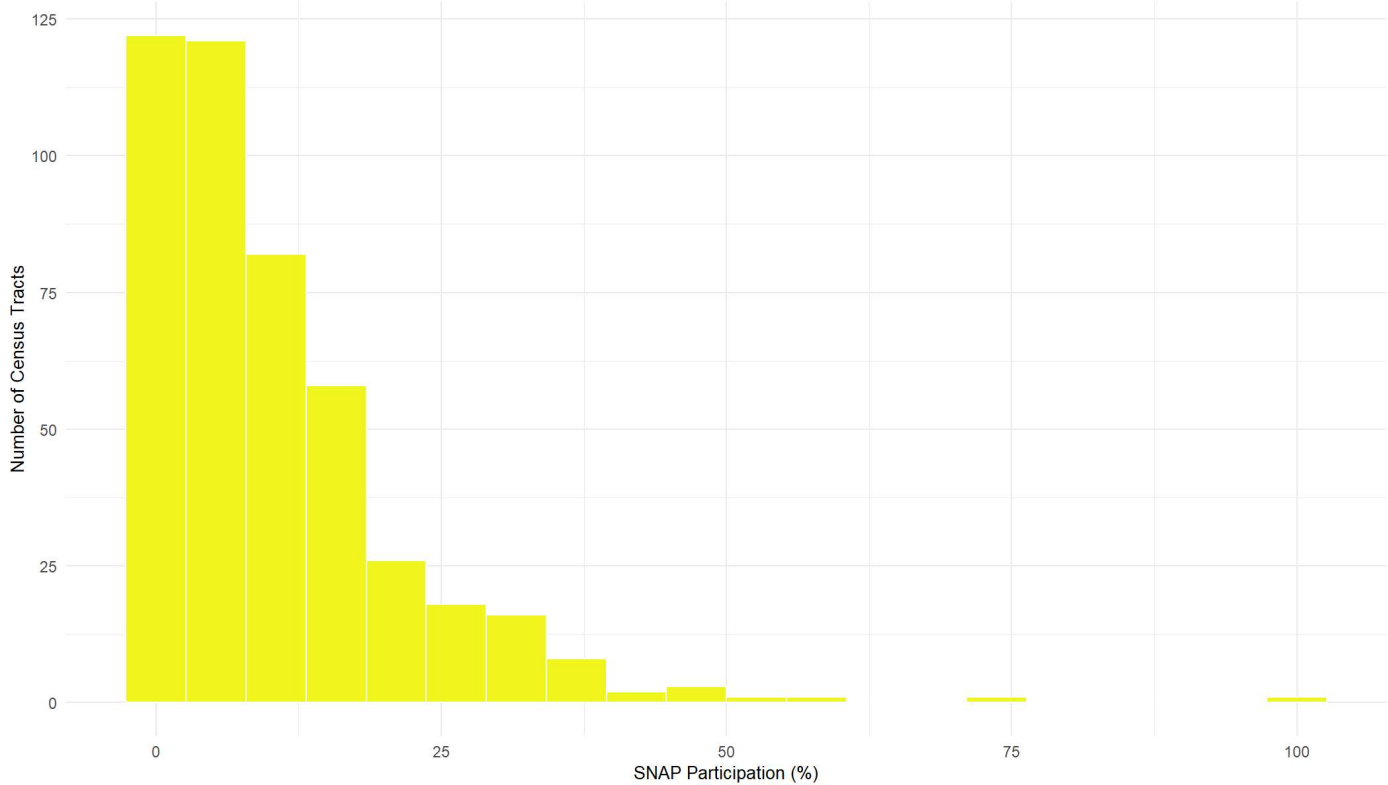
Hints & Resources: - Use `ggplot()` with `geom_histogram()` - Consider bin width - try `bins = 20` or `binwidth = 2` - Add meaningful labels - What does the shape tell you? Is it normally distributed?

```
library(ggplot2)

ggplot(hi_acs, aes(x = snap_pct)) +
  geom_histogram(
    bins = 20,
    fill = viridis::viridis(1, option = "C", direction = -1),
    color = "white"
  ) +
  labs(
    title = "Distribution of SNAP Participation Across Hawai'i Census Tracts",
    subtitle = "Showing variation in the percentage of households receiving SNAP benefits (2023)",
    x = "SNAP Participation (%)",
    y = "Number of Census Tracts",
    caption = "Source: U.S. Census Bureau, 2023 ACS 5-Year Estimates"
  ) +
  theme_minimal(base_family = "Helvetica") +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    plot.subtitle = element_text(size = 12, color = "gray30"),
    plot.caption = element_text(size = 9, color = "gray40", hjust = 0),
    axis.title = element_text(size = 10)
  )
```

Distribution of SNAP Participation Across Hawai'i Census Tracts

Showing variation in the percentage of households receiving SNAP benefits (2023)



Source: U.S. Census Bureau, 2023 ACS 5-Year Estimates

Interpretation Questions

🌲 Part 3: Machine Learning Setup

Now we'll prepare for predictive modeling. First, create the ML-ready dataset:

▼ Code

```
hi_acs_data_ml <- hi_acs %>%  
  st_drop_geometry() %>%  
  select(  
    median_income, poverty_rate_pct, uninsured_u65_pct,  
    unemployment_rate_pct, broadband_pct, renter_pct, no_vehicle_pct,  
    snap_pct, bachelors_or_higher_pct, commute_30plus_pct  
  ) %>%  
  drop_na()
```

🎯 CHALLENGE 4: Build a Random Forest Model

Your Task

Train a Random Forest to predict SNAP participation rates using the other socioeconomic indicators.

Your Model Should: - Use `snap_pct` as the target variable (left side of `~`) - Use all other variables as predictors (`.` on right side) - Have `ntree = 500` for stability - Be named `rf`

Hints & Resources: - Use the `randomForest()` function - Formula syntax: `target ~ .` predicts target using all other columns - Structure: `randomForest(formula, data = ..., ntree = 500)` - Print the model to see performance metrics!

```
library(randomForest)

rf <- randomForest(
  snap_pct ~ .,
  data = hi_acs_data_ml,
  ntree = 500,
  importance = TRUE
)

print(rf)
```

Call:

```
randomForest(formula = snap_pct ~ ., data = hi_acs_data_ml, ntree = 500,      importance = TRUE)
      Type of random forest: regression
      Number of trees: 500
No. of variables tried at each split: 3

      Mean of squared residuals: 45.26575
      % Var explained: 59.1
```

 Understanding Your Model Output **Part 4: Explainable AI with SHAP**

SHAP (SHapley Additive exPlanations) values explain *why* the model makes specific predictions. They break down each prediction into individual contributions from each feature.

Understanding SHAP

Before we calculate, let's understand what we're doing:

1. **Unify the model:** Convert the Random Forest into a format that SHAP can work with
2. **Calculate SHAP values:** Compute explanations for a subset of observations (computing for all can be slow)
3. **Interpret:** Use the results to explain individual predictions and global patterns



CHALLENGE 8: Prepare SHAP Calculations

Your Task

Prepare the model for SHAP analysis by:

1. **Unifying** your Random Forest model using `unify()`
2. **Calculating SHAP values** for the first 200 observations using `treeshap()`

Hints & Resources:

- Step 1: Use `unify(your_model, your_data)` to convert the RF
 - First argument: your Random Forest object (named `rf`)
 - Second argument: your ML dataset (named `hi_acs_data_ml`)
 - Assign result to `unified`
- Step 2: Use `treeshap(unified_object, data_subset, verbose = 0)`
 - First argument: your unified model
 - Second argument: subset of data - use `hi_acs_data_ml[1:200, ]` to get first 200 rows
 - Third argument: `verbose = 0` suppresses progress messages
 - Assign result to `treeshap1`

▼ Code

```
unified <- unify(rf, hi_acs_data_ml)
treeshap1 <- treeshap(unified, hi_acs_data_ml[1:200, ], verbose = 0)
treeshap1
```

	median_income	poverty_rate_pct	uninsured_u65_pct	unemployment_rate_pct
1	1.29346491	-0.09350690	-0.430155482	0.008549443
2	-1.38380920	-2.35406233	-0.260898057	0.023060415
3	-1.04750660	-2.33174656	-0.351068664	-0.675504097
4	-1.31847615	-1.34141248	0.550937545	-0.249404239
5	1.08083242	1.24738495	-0.135753043	-0.080087490
6	-1.39269710	-1.64055473	0.890485451	-0.209096443
7	1.25282692	-2.43469125	-0.199959279	-0.239743266
8	-0.51239473	0.45212917	-0.246519841	-0.345137746
9	-0.13392203	0.49535868	-0.206593791	-1.385637123
10	-2.46863623	-1.33537244	-0.258466107	-0.307488506
11	-1.59217307	-2.26166464	0.074613693	-0.175325946
12	0.96958406	3.97934625	-0.356279143	-0.494097188
13	0.96851999	2.28362398	-0.163215825	1.234986897
14	-0.74633055	-1.96770466	-0.127584792	-0.317592313
15	-0.76053246	-2.10676300	-0.650474840	-1.223138609
16	0.20543581	-0.17987253	-0.432174985	-0.450308612
17	2.24391005	5.78818937	-0.370250790	0.647170373
18	-1.30781674	-1.55897289	-0.204563831	-0.044077859
19	0.56330752	0.50406463	-0.327964922	-0.738218292
20	-0.19616049	0.35267038	-0.207319558	0.163979595
21	-0.20158947	2.50892688	-0.221355821	0.353011991
22	-1.26788588	-1.63728340	1.308669524	-0.398778496

23	0.96715594	7.19839826	-0.320518510	0.622938886
24	-0.19452660	0.16432777	-0.310239643	-0.546411130
25	-0.72542676	-1.99318732	-0.369191075	0.666407018
26	-0.51106304	-1.96303712	0.394716446	-0.373292158
27	1.73346968	2.82449120	0.757226571	1.519237020
28	1.29861693	3.83097873	-0.042980053	0.603970170
29	-0.38822198	1.65988115	-0.042706304	-0.088896439
30	-0.97013850	-2.14681399	-0.014663621	-0.202216554
31	-2.01118707	-1.96975112	0.343457490	-0.305521277
32	4.44765483	2.10865867	-0.141954967	-0.111601162
33	-2.28287501	-2.23213425	-0.297471039	-0.408029468
34	-0.22063184	0.40726783	-0.015229029	-0.274210996
35	-2.31801619	-1.85536005	-0.197053350	-0.202899602
36	4.68842307	5.42715009	0.500166461	-0.296931835
37	1.03316900	-1.12723257	0.433024615	-0.301613237
38	0.23743463	3.27499411	-0.224412302	0.144522782
39	-1.34227480	-2.80230197	-0.203872072	0.374255604
40	1.23625735	6.08694227	0.724577694	1.458772546
41	0.63453787	2.02413067	0.225338826	0.890444990
42	-1.53437868	-0.25700615	-0.303064311	-0.565972634
43	0.97854434	0.69814346	-0.187893555	-0.508294450
44	1.72484857	1.06137742	-0.147641654	0.632518836
45	-0.15491249	0.28609132	-0.133875010	0.637874426
46	-0.48476390	-1.07597317	-0.265932643	-0.325392021
47	-2.01790183	0.46010644	0.196576495	-0.384667581
48	-0.84390436	-1.61103768	-0.251143828	0.444063000
49	1.02981346	5.37645558	-0.030678734	0.075146936
50	-0.08963228	-1.50678668	0.090781664	1.393291961
51	-0.78188843	-2.97799662	-0.405815452	-0.106788690
52	0.05928789	-0.01750869	0.259738489	1.295956664
53	-1.14429956	-0.84569620	0.165772753	-0.136425077
54	-0.83390603	0.52021898	-0.017952261	0.504490031
55	-0.91404839	-1.65773674	0.492870817	-0.054696290
56	-0.71287430	0.32887647	-0.228945248	-0.692112053
57	-0.94912112	-1.89416177	-0.259166336	-0.936408148
58	-1.96144191	-1.73047605	0.427197164	-0.202149590
59	0.79562853	5.68426802	0.881613483	-0.973464086
60	-2.92172712	3.82832059	-0.346521035	-0.902790464
61	-0.52057871	-2.30201109	-0.286098199	-0.060208233
62	-0.65423182	1.32674967	-0.104886633	-0.653730367
63	-1.70030409	-2.70681482	-0.400018092	-0.760481575
64	3.90413984	0.56590749	-0.485530158	0.601656674
65	-0.39479902	2.63462275	-0.016891413	-0.239630483
66	-0.95113349	-1.68274298	-0.420246209	-0.545765260
67	-0.79985560	0.79231324	-0.044483883	-0.093914267
68	-0.36874618	-2.26866063	-0.381290089	0.469678720
69	-1.94126182	-1.97767068	-0.163718779	-0.221142506
70	2.21661681	2.20462398	-0.212066607	1.637742281
71	-0.69335997	0.16273815	0.005907946	-1.008457026
72	-0.36782109	-0.16329851	-0.217700078	-0.475620191
73	2.89057859	-2.11255781	-0.383950681	-0.653005778

74	-1.23223807	-2.00887328	-0.271568831	-0.202320218
75	-1.25853756	-0.82139890	0.022401503	-0.862416488
76	0.34446250	1.03864261	-0.051474750	-0.831691554
77	6.93735549	8.22290660	-0.471191697	0.618894316
78	9.78901732	12.76479820	1.641272023	2.395123364
79	2.09949424	0.87157664	0.796334083	1.138927055
80	-1.18531268	-1.75407259	-0.258109357	-0.435416385
81	-1.09868132	0.72075295	0.142214987	0.337282636
82	0.27300417	-0.82665501	-0.033044882	1.729940356
83	-2.04094642	-1.72806702	-0.187334902	-0.340394926
84	-0.02836254	-1.90576137	0.852096651	-0.202563328
85	0.66425419	-1.58778096	-0.413647287	-0.285012503
86	-0.58870437	-2.02001645	-0.360088743	0.374164463
87	-0.84838797	-1.90391884	-0.111458864	0.801269792
88	-1.54405972	-0.77367640	0.118158588	-0.738795517
89	-0.90959002	-1.77297676	-0.101705590	-0.456816075
90	-1.20448911	-0.69427079	-0.099964944	-0.343007551
91	-1.76364871	-2.24594876	-0.290062409	-0.140521719
92	5.77643839	1.05098209	-0.122370883	1.483452616
93	0.61136735	-2.03975578	-0.273167027	-0.099955300
94	-0.97222070	0.65100676	0.307989579	-0.708205302
95	-0.53657491	-2.50451277	0.087890903	-0.309879154
96	-0.74583554	-1.29637627	-0.342062336	-0.561300956
97	0.59035541	-2.54759114	-0.394378552	-0.723120833
98	-1.06048225	-2.59607594	0.914093983	-0.157376861
99	-2.27789558	-2.16279278	-0.052408191	-0.298999212
100	-0.59979613	-0.09890933	0.503179415	-0.591222130
101	-0.38442498	-0.52209505	0.075520849	0.345329016
102	-0.65685413	-1.61237665	0.320377266	-0.781942127
103	-0.71880583	-1.88975989	-0.497106972	0.744303035
104	0.81046926	-2.51148790	-0.348971671	0.303906210
105	-0.38491104	2.29232239	0.154350569	0.652097284
106	1.03093455	4.42301681	-0.638096182	0.603474709
107	-1.07796645	-2.18520721	-0.387448858	0.100949101
108	-1.08697125	-1.87716899	-0.264388593	-0.325273333
109	-1.01005261	-1.51721324	-0.422489951	-1.089705767
110	1.39900068	0.08729734	0.092533927	-0.004808931
111	-0.59297499	0.80789825	1.065977230	0.377492343
112	9.26985499	8.66854894	1.574975979	-0.244555968
113	0.12889793	0.10641690	-0.287719612	-0.730752151
114	-0.38557873	-2.05524455	0.010104133	0.995552519
115	-1.06215047	0.10922765	0.295339849	0.479093307
116	-1.39304102	-2.48813927	-0.200936454	-0.363498338
117	1.84959152	-0.48460448	-0.262801182	-0.630019085
118	-1.10228842	-1.39739720	-0.248091009	-0.468749074
119	0.44828136	1.26805584	-0.268313583	-0.737708708
120	-0.55989131	-0.85481068	1.349079860	0.996182049
121	0.60002504	0.48330009	-0.103424032	0.728203655
122	-0.95154747	0.05844084	-0.156551110	-0.446961721
123	-0.72651140	-0.60825866	-0.378023864	-1.060412845
124	-1.19336255	-0.17392355	-0.012928145	-0.624401017

125	0.17586106	-0.97017848	-0.113704710	-0.713900181
126	-1.04634576	-2.24537834	-0.483072200	0.253958410
127	-1.51709133	-0.33086210	0.389769906	-0.723086025
128	-1.67061792	-0.74205530	0.096933139	-0.104238713
129	-1.97110026	-1.50686303	-0.182772821	-0.239149357
130	-1.11228553	-4.86174177	-0.630495521	-0.006135545
131	1.02168622	1.56576253	-0.208485187	0.704004419
132	-2.04448342	-2.13710063	-0.211605602	-0.096124239
133	-0.53028564	0.26752173	0.106917757	-0.054715288
134	-0.41357515	-2.46758297	-0.482562834	-0.748613706
135	-1.29055111	-2.38805632	-0.683373675	-0.408591900
136	0.02495109	0.86063286	0.786485866	0.114861344
137	6.91454174	0.28566855	-0.462498633	0.603375255
138	5.12075377	5.92301454	-0.431976129	-0.247720898
139	-1.52067641	-1.56409116	0.262586878	-0.848681360
140	8.60083194	7.48468379	0.429530754	0.814877807
141	5.95435570	4.75141367	-0.252641566	-0.327512940
142	-0.17653042	-0.95575433	1.741244591	0.838992542
143	-1.79458024	-1.65015338	-0.323034970	-0.584073790
144	-0.62207071	-0.55821956	-0.183311559	-0.077387055
145	-2.07928456	-1.23547635	-0.192754470	-0.506488279
146	-2.13275697	-2.23622235	-0.293927034	-0.297073796
147	-0.56182280	-0.35089807	-0.356679810	0.130747508
148	-1.75817968	-1.86033507	0.084642012	-0.321473173
149	-1.48089824	-0.03638486	-0.355468501	-0.074867587
150	-0.58651761	0.23052310	0.295711061	-0.065878415
151	-0.68590698	-1.55524054	-0.252237530	-0.137980940
152	0.94113030	5.66733022	0.535123375	0.788335875
153	3.15852880	2.16842662	-0.014416805	0.315909881
154	0.11106554	1.36614661	0.260870839	-0.118921256
155	-0.65080110	1.11552784	0.084976316	-1.485253265
156	2.49072771	0.99414190	-0.040100698	0.789415026
157	-1.01505327	0.52397563	0.146432779	-0.612595226
158	3.35719429	4.73014002	-0.318391969	0.486248627
159	4.14580015	4.98173551	0.952121607	-0.007230760
160	4.38988371	7.11506288	0.241950293	1.350650863
161	-1.83101871	-1.44075011	-0.149835620	-0.224023731
162	-0.64629586	-1.30631270	-0.180228075	-0.051290631
163	0.30939658	-0.21113186	-0.470076776	-1.329526585
164	-1.19604513	-3.30943998	-0.589112731	-0.096026751
165	0.50143651	-0.04663692	-0.315598505	0.042360149
166	-0.65767096	0.25558755	0.447834095	-0.655315081
167	-0.59108008	-2.94220715	-0.494393893	-0.207818644
168	-0.85420790	4.28608293	0.022476459	0.653745390
169	-1.03689142	-2.79682032	-0.439985279	0.687082001
170	-1.87321898	1.43198272	-0.042791362	0.696113731
171	-0.97383564	0.88629257	0.778292362	0.630047397
172	-1.13760010	-2.08575655	1.025582095	-0.190582332
173	4.38928044	-3.59456526	-0.521757207	-0.730125432
174	-1.88337440	-2.20782043	0.598266186	0.526101671
175	-1.76056426	0.87826007	-0.378123115	0.229343622

176	4.19391066	6.49914521	0.048601362	-1.804577476
177	-0.69182363	-0.19447945	-0.198298199	-0.080360339
178	0.42232144	-3.39417740	-0.580435778	0.426045527
179	-1.04713869	-2.36705865	-0.173937233	-0.497300025
180	-0.82091566	0.87331400	-0.143746666	-0.319951994
181	-1.00405123	-2.24680583	0.243377252	-0.690026601
182	5.93919986	6.08638544	0.991898583	2.226645108
183	-1.11219095	-1.87927242	-0.473460642	0.054216827
184	-0.92784123	4.68170192	0.224208672	-0.148058856
185	2.30649230	6.36313621	1.755267368	2.707447345
186	1.35671823	2.23333848	-0.096894646	0.690985011
187	-1.58877318	-2.42164828	0.303525657	-0.305649609
188	-1.35238414	-1.91295603	-0.308605555	-0.089958352
189	-2.20153291	4.90255453	-0.441422983	1.098362126
190	-0.34804607	-2.01239443	0.742519223	-0.804201748
191	-0.78660742	1.07287565	-0.066499252	-0.313784550
192	-0.52567297	-1.29596936	0.534693700	-0.557942437
193	0.81135002	0.79955255	0.046562180	0.249565006
194	-0.32788818	-1.50771308	0.147926771	-0.091490185
195	-0.94302588	-0.35914256	-0.089795931	-0.418754362
196	0.09530692	-1.31303595	-0.566426121	0.353346741
197	0.42482913	1.23067716	0.070860571	0.514101751
198	4.50812630	-0.10454150	-0.252935638	0.345635163
199	-0.82936970	-1.84572998	-0.456517494	-0.264386938
200	-0.87946824	-0.27807447	0.667236157	0.506373587

	broadband_pct	renter_pct	no_vehicle_pct	bachelors_or_higher_pct
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1	0.97920431	-0.375728940	-0.7637894828	-4.43504010
2	-1.03753694	0.091855297	-0.0671959525	-2.98245135
3	1.01502506	0.045052762	0.0546998599	0.20159811
4	-0.54875187	0.199758913	-0.0870284037	3.37352162
5	0.61371955	0.107256671	0.3268025614	-1.34580434
6	2.62777547	0.062916880	-0.0326351272	6.05691141
7	2.19085441	0.434846024	0.2104394706	2.54719729
8	-0.57726186	0.094827860	0.0734937491	-0.40304845
9	-1.61876960	-0.060728201	0.0456387573	-1.06945211
10	-1.16320918	-0.436057021	-0.2040328781	-3.69261532
11	2.46386323	-0.210028875	-0.0844130155	3.01781964
12	-0.05803681	-0.142549341	-0.1811676752	-3.25634784
13	1.97996928	0.116175895	0.3905020171	1.86897065
14	-0.48281322	0.131874869	-0.0007878037	-3.17050199
15	-0.64851698	-0.021942076	0.1862545499	-1.25406682
16	1.16785538	-0.436435248	-0.6113187114	0.14184130
17	1.46178411	0.250557562	0.4445783145	10.07306169
18	-0.75778048	0.161155326	0.1629576414	-2.30170132
19	1.05093687	-0.087589430	-0.5207790401	-2.20013777
20	1.92713182	0.248811456	-0.2741976507	-3.90540057
21	1.72520780	0.049026239	-0.2975099078	-4.25202163
22	-1.04315723	0.100267611	0.3112076466	-1.03721234
23	-1.70701868	-0.270214531	0.2352103385	-0.26670493
24	-0.86595778	-0.270991968	-0.5519228981	-4.14773788
25	-0.38178483	0.102278780	0.0337611646	-3.30193246

26	0.93042338	-0.010250920	-0.0826670639	1.08957363
27	2.29295633	-0.168569848	0.4584672936	-0.37971056
28	-0.60750470	-0.228275830	-0.2596778718	4.09931568
29	2.18257399	-0.153311742	-0.2679019434	2.44019987
30	2.85024602	0.460977769	0.2928050349	3.86454436
31	-0.22520281	-0.007506289	0.2221456150	-3.70508931
32	1.87338813	0.321560865	0.0911863227	12.39457611
33	-0.27956547	0.102951281	-0.1160401864	6.13662913
34	-0.12834753	0.238824840	0.0511154025	-3.45386210
35	-1.15404540	0.005795820	-0.1622259281	-3.21650159
36	3.10906451	0.485476666	0.3230164521	4.43330149
37	1.99085045	0.077543043	0.1250338479	2.04393494
38	0.10013938	0.164256842	0.1328983819	0.58735485
39	-0.45620459	0.360924062	0.0286063575	5.35252700
40	2.50254579	0.360381953	0.0172320518	3.71888969
41	1.61784931	-0.376215560	-0.4476908688	-4.46306164
42	-1.96441728	-0.161115216	-0.5084140038	-0.26384922
43	-0.47084134	0.160411909	0.4390380522	-2.87810436
44	0.15250722	0.086827937	-0.0521278938	-2.73303471
45	0.65695474	0.487161524	0.3059432293	0.59838756
46	-0.14112262	0.046564754	-0.3369192803	-2.98256863
47	1.61501910	-0.521982223	0.1607567147	8.97739922
48	0.22022968	0.368935145	0.1519168024	-3.06054637
49	3.80479527	0.392245651	0.5146030903	3.17429124
50	0.14680337	0.344578080	0.1357475825	-0.89111057
51	-3.77377708	-1.257005155	-0.0076841139	-0.21869777
52	-1.48101045	0.454270076	0.2987917575	-0.92916163
53	-1.12848607	-0.086604980	0.0217528005	-2.95134244
54	-1.62574986	0.187234342	0.2235209419	0.25582792
55	-1.20944986	0.366550789	0.2137983754	-0.22444198
56	-1.42905630	-0.064581014	-0.1299038837	-3.48573732
57	-0.16384025	0.360924217	-0.0171312806	0.24708855
58	-0.89603563	0.146783020	0.2917097444	-3.01033015
59	2.49094728	0.180540797	0.1590859590	0.07083985
60	-2.00105345	-0.171606463	-0.2111968787	-3.75849953
61	2.30517028	0.027503579	-0.1356994809	1.42551973
62	-1.11292831	-0.381562379	-0.1510925030	-4.01040467
63	-3.16774381	-0.039474377	-0.0198546344	0.02813579
64	1.63223312	-0.250312411	0.0168523675	-2.78841989
65	-0.09848439	0.299918484	0.0544963494	-2.95373008
66	0.41046675	0.026914540	-0.1231083155	-2.23912410
67	-0.41352710	-0.006794811	0.1500176622	7.41062630
68	-2.02782688	-0.576351541	0.0514697999	-1.89407588
69	-1.09401018	-0.124055493	-0.0081712627	-3.19255934
70	2.24800792	0.369642052	0.0496698433	-3.03122600
71	-3.09010306	0.326820864	-0.1527918953	-0.09908383
72	-0.61698100	0.099169609	-0.0027395947	0.24439851
73	1.47729436	-0.219341659	-0.4471554287	-3.67783011
74	-1.03781379	-0.399293389	-0.1774074643	-3.20580483
75	-1.37778984	0.028847180	0.0521327278	0.42094423
76	-0.62034131	0.017540458	0.0548425173	0.60766801

77	1.76736222	0.367612948	0.1819227316	-1.29291978
78	2.11664281	1.915826443	0.2461659909	14.90626528
79	0.12988051	0.571596667	-0.1454967003	12.43845386
80	-0.76821291	-0.105459700	-0.2039228210	-3.74273001
81	-0.92069015	0.037037835	0.0749283725	2.76113706
82	2.93304707	0.454237433	0.6687714384	2.67637217
83	-1.56129297	-0.146331046	0.1737108907	-2.56024533
84	0.57122658	0.224810666	0.1359126504	1.57407174
85	1.00276922	-0.324356649	0.0976021537	-1.29615251
86	-0.20157151	0.246938047	0.1572363323	-2.16700001
87	-1.53967743	0.003945590	-0.0612981377	-1.30073048
88	-1.84862877	0.150373688	-0.2125893238	0.95032992
89	-0.33665109	0.208446684	-0.0227149919	-3.05609928
90	1.71397260	0.307367910	0.3184193568	1.99226269
91	-0.68615128	-0.416877149	-0.0359604048	2.74861707
92	-0.17775331	0.281598822	0.0023816091	11.43537315
93	-1.47684115	0.362465827	0.0771051707	0.59919986
94	-0.46558630	-0.030058846	-0.0884321730	-1.43434765
95	0.64929475	-0.352546995	-0.2025757574	-1.19546855
96	-0.12676211	0.533369816	-0.0099588990	0.44099365
97	-0.81655154	-0.352498285	-0.0276430344	-2.92194324
98	-1.28454249	0.243607606	0.1299568594	4.72423443
99	-1.22630371	-0.185304026	-0.0713409399	-3.57017300
100	-0.60585803	-0.022408683	-0.0968076393	-2.11074289
101	-0.99478170	0.250794576	-0.0083128338	1.27572989
102	-1.05102230	0.107366743	0.0269515940	0.88928475
103	-0.23866690	0.223522622	0.1377941373	0.62194275
104	-2.03804868	-0.907961286	-0.0165867680	-2.29005232
105	-0.05169887	0.337809114	0.3837131592	1.97534025
106	0.87011486	-0.777234544	-1.1707678358	-4.49910017
107	-0.32053724	0.049536929	-0.1347298717	6.37843731
108	1.67487032	0.411685849	-0.2896936416	-0.80709595
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110	3.16647484	0.039548624	0.3256112327	9.10755955
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112	4.64138912	1.479810723	0.3754954732	6.65400735
113	0.70020545	-0.223912566	-0.5922188464	-4.06139618
114	-1.69218472	-0.317907017	0.1033416451	-1.89387121
115	0.80483976	-0.008339036	0.0983768119	-0.25650968
116	-1.38246058	-0.244113219	-0.1445694139	-2.08737238
117	-0.18003704	-0.124763988	0.2148919095	-3.60996021
118	-1.00379858	-0.040410339	0.2123878506	-3.12738627
119	-3.09770213	-1.386424611	0.0066998735	-2.47424865
120	0.63404328	0.567533162	0.1837952774	0.26234793
121	2.12195767	0.465281324	0.4209766335	-0.32601459
122	0.90402598	0.017754440	0.1318236811	5.17173653
123	1.29631553	-0.248672202	-0.5315285789	-1.98963604
124	-0.12441343	-0.144963578	-0.2200128898	1.31272795
125	1.63484723	-0.094348301	-0.1326682919	-0.97419027
126	-2.30015121	-0.721512924	0.0458191833	-2.30015923
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128	-0.25646627	0.005652120	0.2743604547	2.04453000
129	-0.83476296	0.168497333	0.0204812800	-2.61617178
130	-3.74310485	-1.025532260	-0.3019587715	6.08661852
131	0.63671934	0.125938668	-0.0707981692	-0.42806087
132	-1.72393041	0.001737323	-0.0959190850	-3.13963216
133	0.96172622	0.140092478	0.1917207869	3.61296077
134	1.21174025	-0.223380702	0.1607630086	-0.69364017
135	-0.74908151	-0.268001522	-0.2846465551	-0.20723021
136	2.39076474	0.780750315	0.2459767695	1.98542196
137	1.79337735	-0.072090981	-0.3833744325	-1.89207371
138	2.78645041	0.271004322	0.2350748139	-1.79581569
139	-0.12402030	0.030201071	0.0052732057	2.04921839
140	4.13037632	1.761263052	0.3154962001	11.29103302
141	2.34693509	0.087829403	-0.0015880408	9.85956948
142	0.24097617	0.472706168	0.2892576477	1.46176878
143	-1.02323157	0.220681985	0.2656944462	-0.95085398
144	-0.66845382	0.184867249	0.0376640442	-3.16521464
145	-1.72964686	-0.061610236	-0.0896665350	-3.06264171
146	-1.28535210	-0.257058178	0.1198137979	-3.19291475
147	-2.26668185	-1.090266904	-0.1467346442	-0.02372005
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149	-1.08078171	-0.093908362	-0.4415834725	-4.31814877
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151	-0.87030815	0.388733029	0.1183804987	1.46953368
152	1.39141475	0.126386806	0.3522136114	6.84586430
153	0.21264874	1.845235219	0.3853725191	-0.53677711
154	0.54585024	0.455243333	0.3802767693	-0.74922424
155	-0.53147194	-0.043907247	-0.0292413143	-1.48323690
156	-0.80804223	0.746300038	0.0477375008	0.05560844
157	-0.84670716	0.058269897	-0.0448348301	-3.88349740
158	1.94215593	-0.130291222	0.0832875871	7.47009353
159	1.17629771	0.385423283	-0.0205377478	11.25570112
160	2.95206741	0.433248691	0.5646465638	9.54052219
161	-0.88938181	-0.064800134	0.0382252798	-2.81081421
162	-0.32887871	0.348669029	0.1155821094	-2.70061759
163	0.76381664	-0.379840711	-0.6296007928	-1.58181833
164	-2.32601240	0.091598463	-0.0588793803	-0.84419477
165	-2.33973111	-1.184030020	0.0148751052	-0.52445449
166	-0.39684868	0.212911753	-0.1347802250	0.82177160
167	-1.71810958	-0.880630211	-0.0823323878	-0.36442165
168	1.88118607	0.182545924	0.3209876929	9.29957596
169	0.08227323	-0.190806485	0.0539833198	-2.81588652
170	-0.41989549	-0.027348482	-0.0334770833	12.59258596
171	-0.56034745	0.235790032	0.0689371914	12.15211343
172	1.23462498	-0.104934919	-0.0591064442	1.48068564
173	-3.94780997	-1.156258439	0.0569926991	-1.08257376
174	-1.03332080	-0.108940645	0.0419888187	-3.39903410
175	0.16392859	-0.523551483	-0.3469968112	8.26422809
176	-0.90387587	0.258589955	-0.0670950466	7.77990136
177	0.88939896	-0.011586626	-0.6117721373	-4.52587475
178	-1.65730074	-1.556791982	-0.0613864402	-0.78529410

179	-1.75323896	0.019720352	-0.0209069212	-2.56477655
180	-0.31959777	-0.126014903	0.2248952835	10.92692856
181	-0.63218637	-0.155908428	-0.2896148009	-3.65792173
182	4.27707389	0.830102242	0.1819144052	11.13449254
183	1.59590009	-0.072510318	-0.2952938663	-3.34721847
184	-0.30474116	0.098467752	0.0323474215	0.40337089
185	2.40919431	0.600206357	0.4267584122	12.95871283
186	-1.76821739	0.057578682	0.2405755564	-1.77764160
187	-0.55739883	-0.119287427	-0.1570372520	-4.00428268
188	-0.17293525	0.223337153	0.1247645504	-0.12424147
189	-1.30049201	-0.320611252	-0.1259441259	7.49152256
190	-1.16997013	0.076096274	-0.0349134034	1.11877621
191	-0.86183213	0.179735919	0.3703223990	-2.96144306
192	-0.93747149	0.002670349	0.0720589703	0.21328826
193	1.12187864	0.205623233	0.2504503843	10.25377834
194	-0.18155852	0.614177281	0.0932128454	0.28893310
195	-0.69685382	-0.061151368	-0.0669213116	-2.75720749
196	-1.46933944	-0.202105767	-0.0673842969	-1.45376592
197	3.05578926	0.565606734	0.4308773832	6.52079656
198	2.11359040	0.233166985	0.3224665384	3.74722848
199	0.88949642	0.086890351	-0.1583313401	-0.08942774
200	-0.17193883	0.389903112	-0.0678715862	0.62127934

commute_30plus_pct

1	-0.8031520727
2	0.0019241335
3	0.0234203578
4	0.0933906612
5	-0.2482616649
6	-0.0611367942
7	-0.6806002144
8	-0.2131775446
9	-0.4517711450
10	-0.0806740813
11	0.0408035123
12	-0.1896826528
13	-0.3857571524
14	-0.0264503083
15	-0.1678637524
16	-0.3512835597
17	0.7199250080
18	0.0822799312
19	-0.6746598479
20	0.0319849881
21	-0.5036525507
22	0.1609652432
23	-0.0468292991
24	-0.3283131491
25	-0.0009400236
26	-0.2500099035
27	-0.5707538611
28	0.1849041876

29	-0.2562086871
30	0.6277547813
31	-0.0921272860
32	-0.1529168087
33	-0.0997848999
34	0.0994911457
35	-0.1011405198
36	0.4644865455
37	-0.6834196386
38	-0.2869682213
39	-0.0174222424
40	-0.4042362624
41	-0.6739300412
42	-0.1470117190
43	-0.1306691567
44	1.6067761361
45	0.1743301974
46	1.2305079916
47	0.4673720309
48	0.0881993890
49	1.0592694774
50	0.2072475001
51	-0.3602728817
52	0.1839215080
53	0.0897255296
54	0.2289568704
55	0.2148458124
56	-0.3665717295
57	-0.0596924135
58	0.0670910503
59	0.2215595429
60	-0.0414543530
61	0.0282359703
62	-0.0733094991
63	-0.2819405740
64	-0.6354254490
65	0.2584973188
66	-0.1440952559
67	-0.1464434542
68	0.0370356724
69	-0.0088184948
70	0.6273323829
71	-0.2952566201
72	-0.0662336471
73	-0.5366043790
74	-0.0886598251
75	0.1973300774
76	-0.0298982966
77	-0.3287456338
78	0.2861982876
79	-0.0678267714

80	-0.1159208546
81	0.0894147433
82	0.2706038607
83	0.1044263366
84	0.1381932795
85	-0.3833196138
86	0.0298906555
87	-0.2103613001
88	0.1558108447
89	0.0602032700
90	0.4152251480
91	-0.0927029258
92	0.4857923479
93	0.4485040134
94	0.0512557398
95	-0.2922437982
96	0.1650122742
97	-0.3035673253
98	0.1483482460
99	-0.1255382497
100	-0.0271043850
101	-0.0631504222
102	-0.0799990719
103	0.1567674881
104	-0.0387282390
105	0.5340928670
106	-0.8972018907
107	-0.0468080048
108	-0.0056675720
109	-0.1049265413
110	0.9694202372
111	0.3645254860
112	-0.0085212029
113	0.4403586367
114	0.0199824491
115	0.1925286611
116	-0.0894430849
117	-0.4560960809
118	-0.2241769163
119	-0.2717215244
120	0.2499186953
121	0.2592347049
122	-0.0621671240
123	-0.0522623800
124	-0.3928184034
125	-0.2802650514
126	-0.3165994679
127	-0.0866141260
128	0.1131807035
129	-0.0246417224
130	0.1903224258

131	0.4411207389
132	-0.0013495703
133	0.1445671898
134	-0.1199039200
135	-0.2409173628
136	0.3019940167
137	0.0443145681
138	-0.1595742934
139	-0.0310406488
140	-0.0178405567
141	-0.0877162364
142	0.4562293975
143	0.1513754403
144	0.1150873361
145	-0.0885707541
146	-0.1039614495
147	0.0774725872
148	-0.0003400179
149	-0.1897847375
150	0.3407692178
151	0.0082662527
152	-0.3823113511
153	3.1823793148
154	0.4333267436
155	-0.3791923618
156	0.4709042847
157	-0.1228996240
158	0.5590016652
159	-0.0456436102
160	0.8885660127
161	0.0112620050
162	0.3107566117
163	-0.8025313481
164	0.0111938043
165	0.5491688274
166	0.0967402147
167	-0.0960611876
168	0.1664964420
169	-0.1204872640
170	0.1756382875
171	-0.0795819742
172	0.1785536164
173	-0.4934248244
174	0.0569594722
175	-0.0693000375
176	-0.0544014403
177	-0.5493265625
178	-0.1099794324
179	-0.0244377211
180	0.4560572176
181	-0.0462247654

182	0.1534300278
183	0.1345201361
184	0.1054061578
185	0.4369151787
186	0.0566207912
187	-0.2420495687
188	0.1683579948
189	-0.1591907135
190	0.0088562124
191	0.3882715015
192	0.0299178589
193	-0.2860460290
194	0.1612890740
195	-0.0194645074
196	-0.4203975462
197	0.6864341724
198	0.7108905174
199	-0.1611853775
200	0.1219677695



Why Only 200 Observations?



Patience Required!

The SHAP calculation might take 30 seconds to a few minutes depending on your computer. The `verbose = 0` parameter keeps it quiet while it works. You'll know it's done when the next code chunk starts executing!



CHALLENGE 5: Feature Importance Plot

Your Task

Create a feature importance plot showing the top 5 most important variables for predicting SNAP participation.

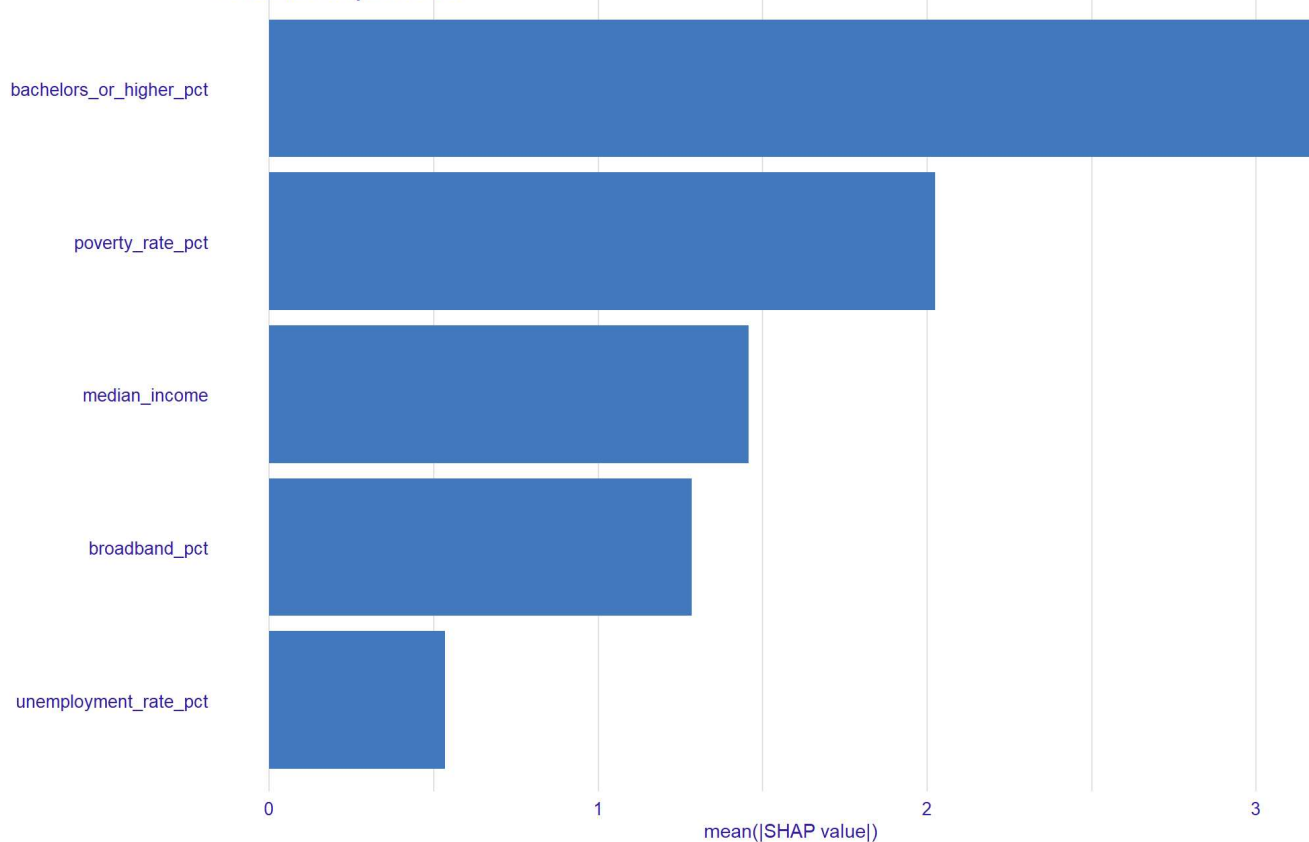
Hints & Resources: - Use `plot_feature_importance(treeshap1, max_vars = 5)` - This uses the SHAP values we just calculated - The plot shows which features have the biggest impact on predictions - Longer bars = more important

▼ Code

```
library(treeshap)

plot_feature_importance(treeshap1, max_vars = 5)
```

Feature Importance

 Interpretation

CHALLENGE 6: Feature Dependence Plots

Your Task

Create feature dependence plots for the **top 5 most important features** you identified in Challenge 5.

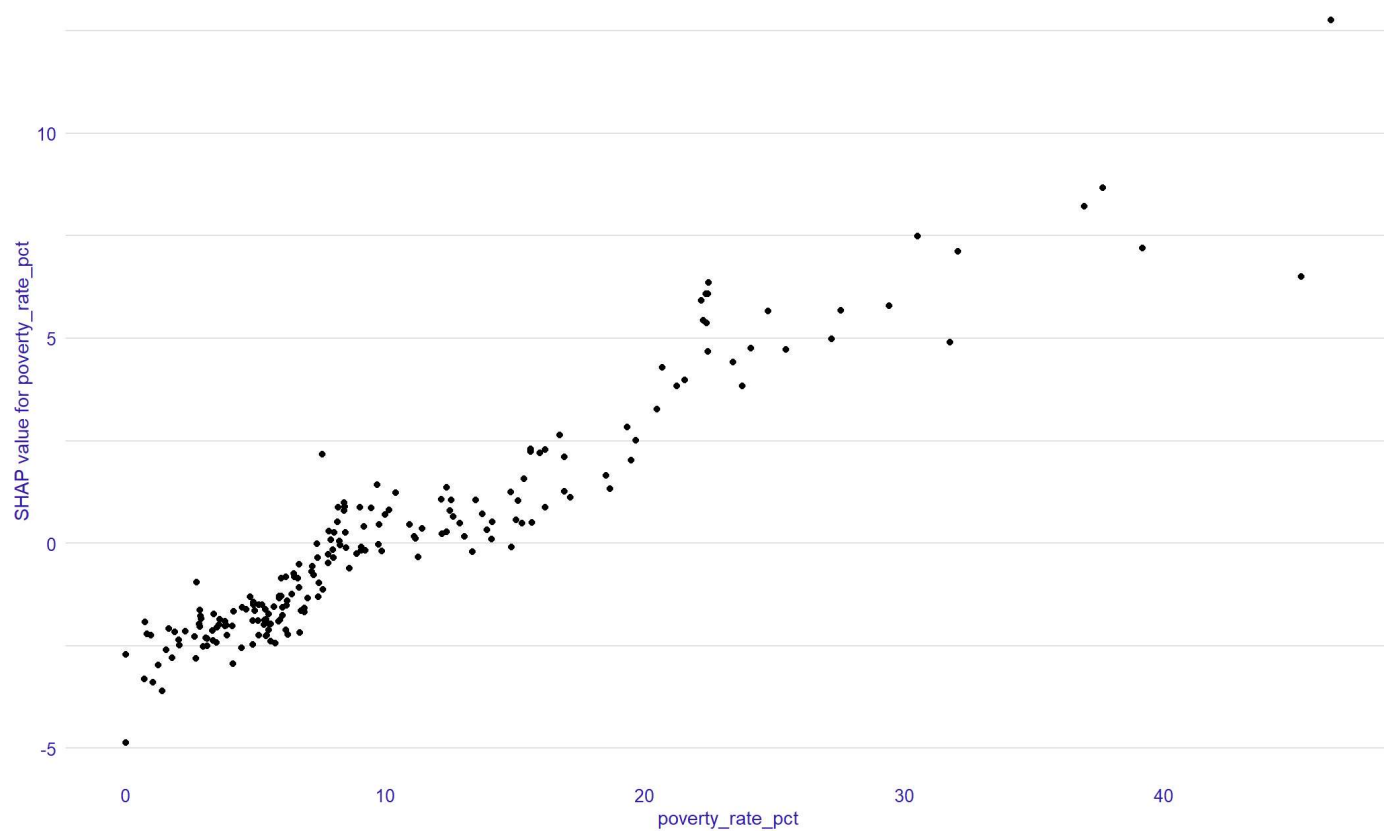
What to Do: 1. Look at your feature importance plot 2. Identify the 5 most important variables 3. Create a dependence plot for each one 4. Use: `plot_feature_dependence(treeshap1, "variable_name")`

Hints & Resources: - The x-axis shows the actual variable values - The y-axis shows SHAP values (how much it pushes predictions up/down) - Red dots = push toward higher SNAP % - Blue dots = push toward lower SNAP % - Look for patterns: linear? non-linear? threshold effects?

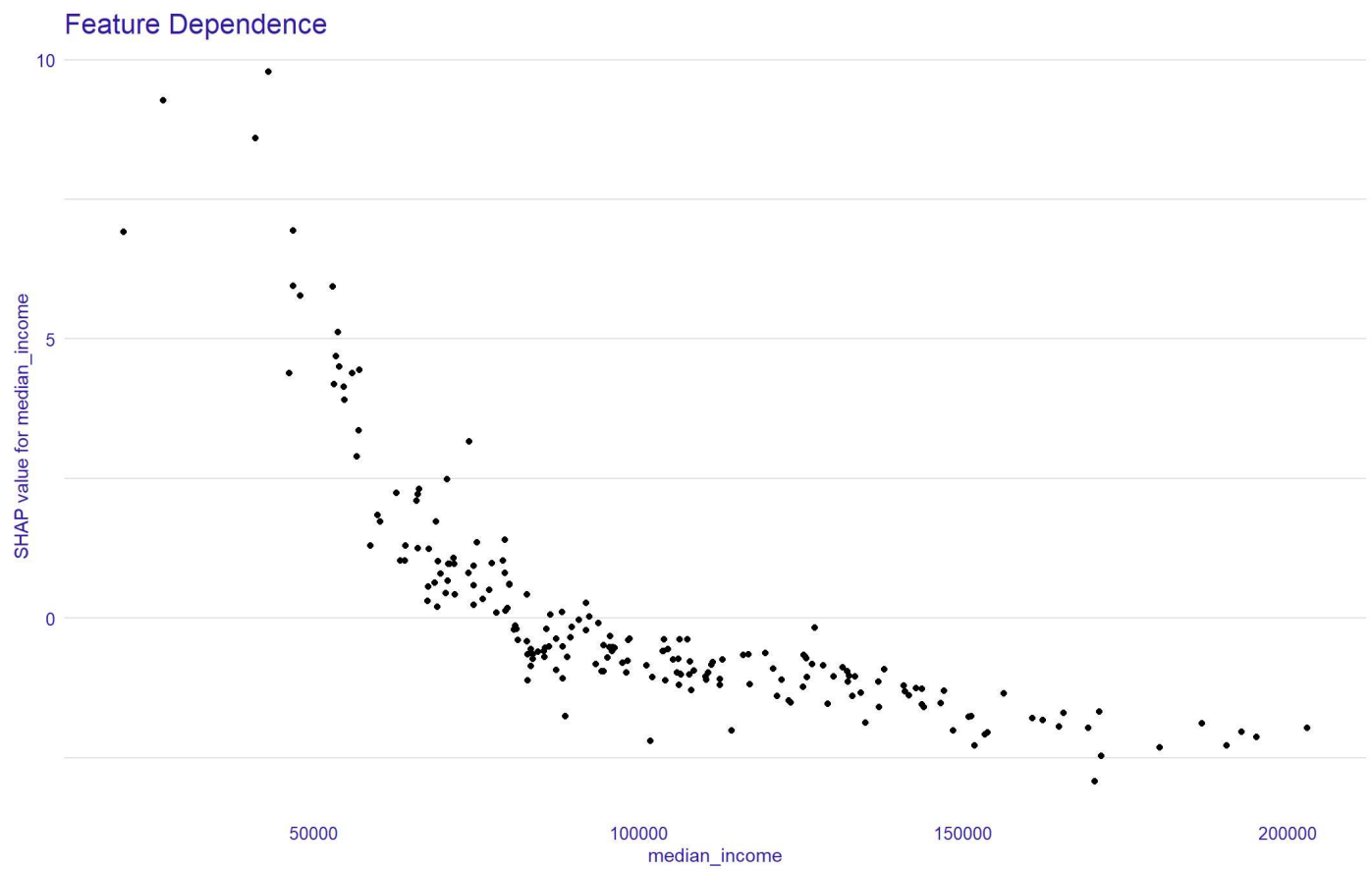
Your Code (plot all 5):

```
plot_feature_dependence(treeshap1, "poverty_rate_pct")
```

Feature Dependence

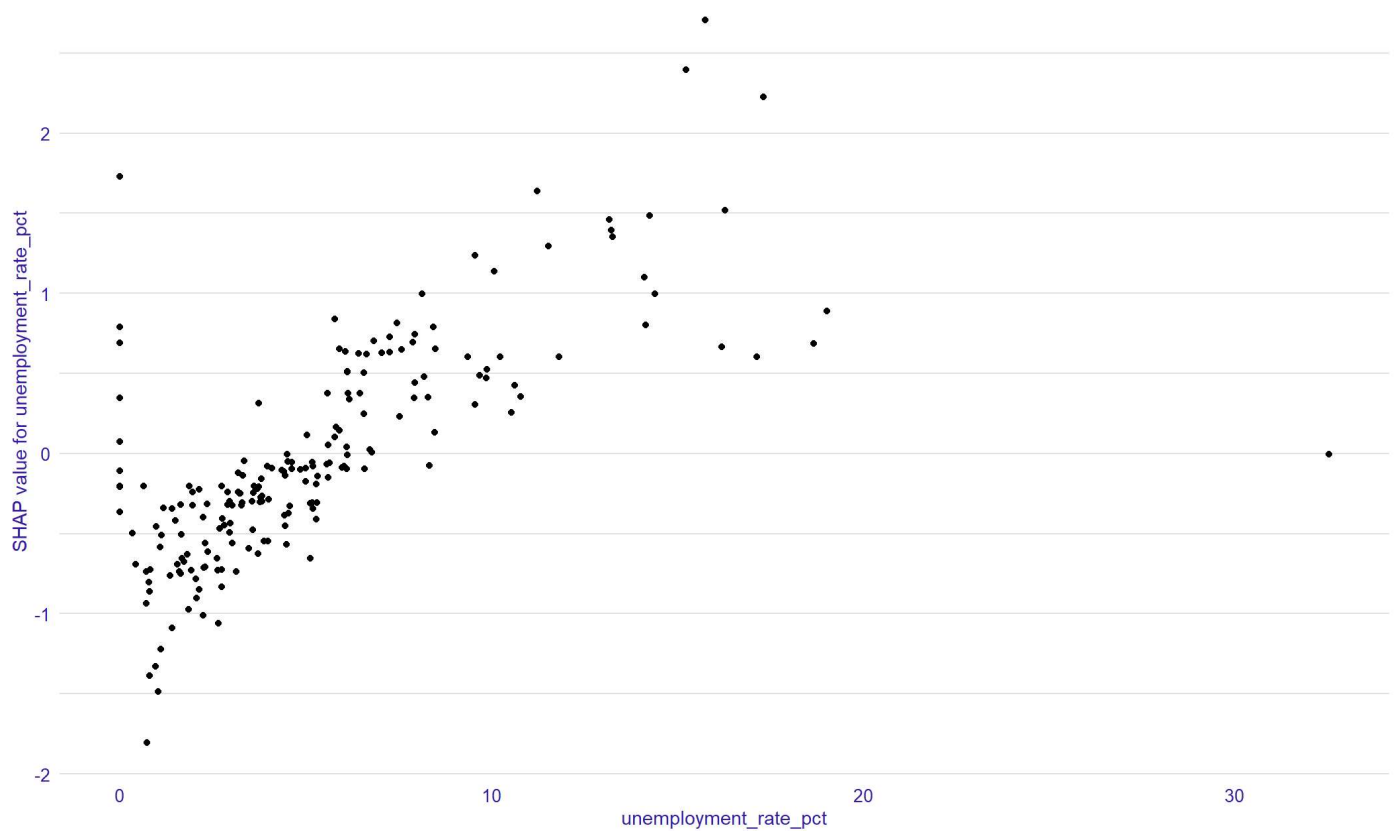


```
plot_feature_dependence(treeshap1, "median_income")
```



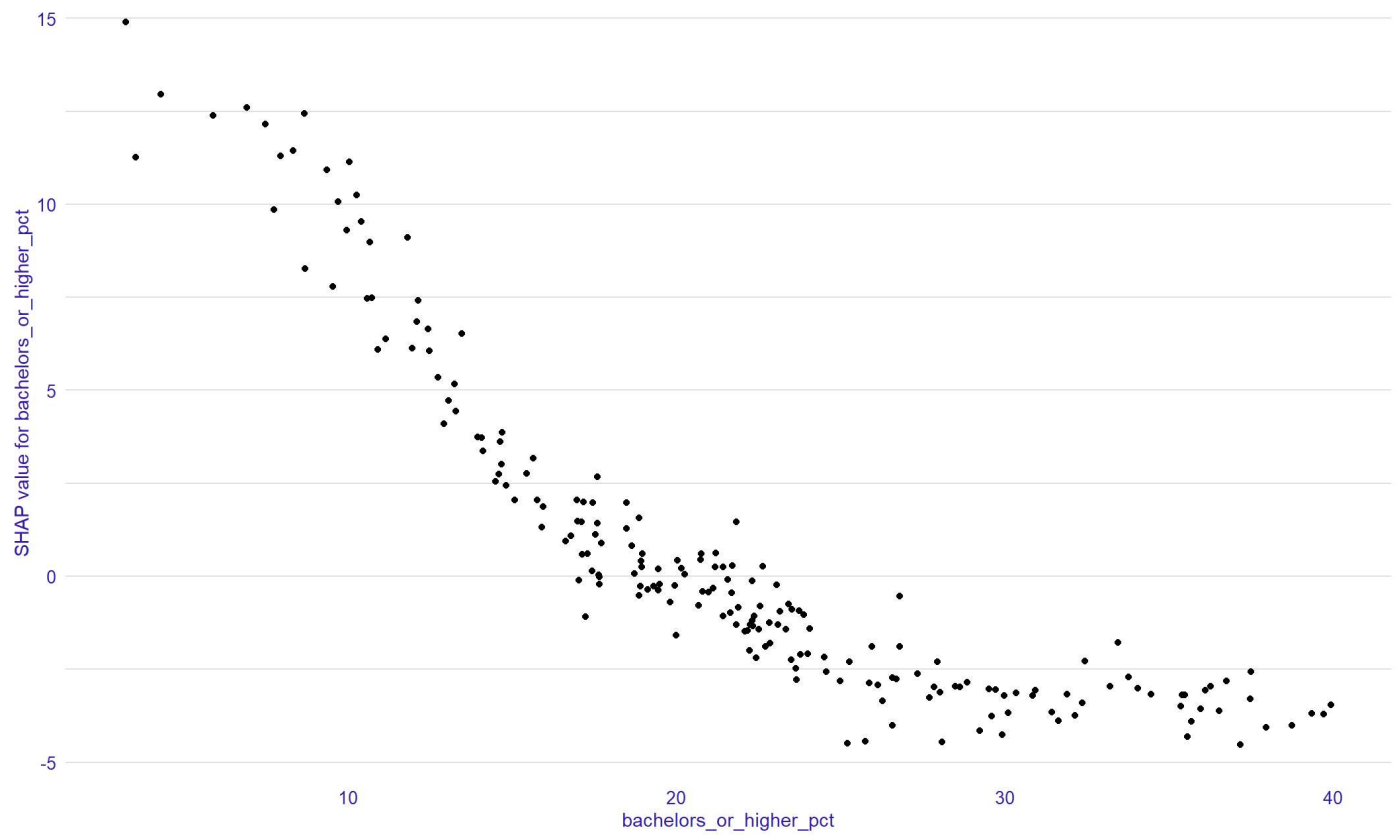
```
plot_feature_dependence(treeshap1, "unemployment_rate_pct")
```

Feature Dependence

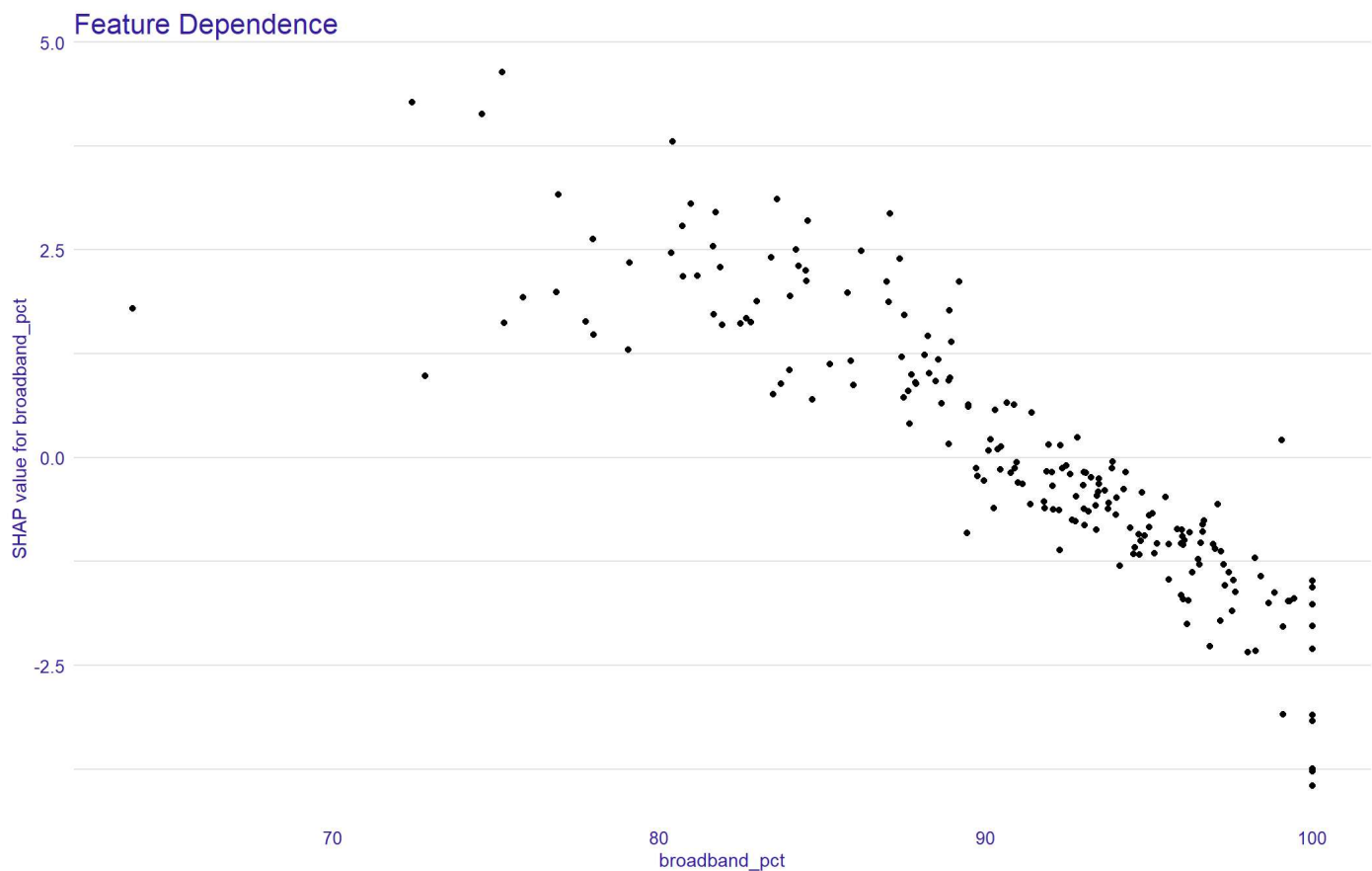


```
plot_feature_dependence(treeshap1, "bachelors_or_higher_pct")
```

Feature Dependence



```
plot_feature_dependence(treeshap1, "broadband_pct")
```



Dependence Analysis Questions



CHALLENGE 7: Individual Tract Contribution

Your Task

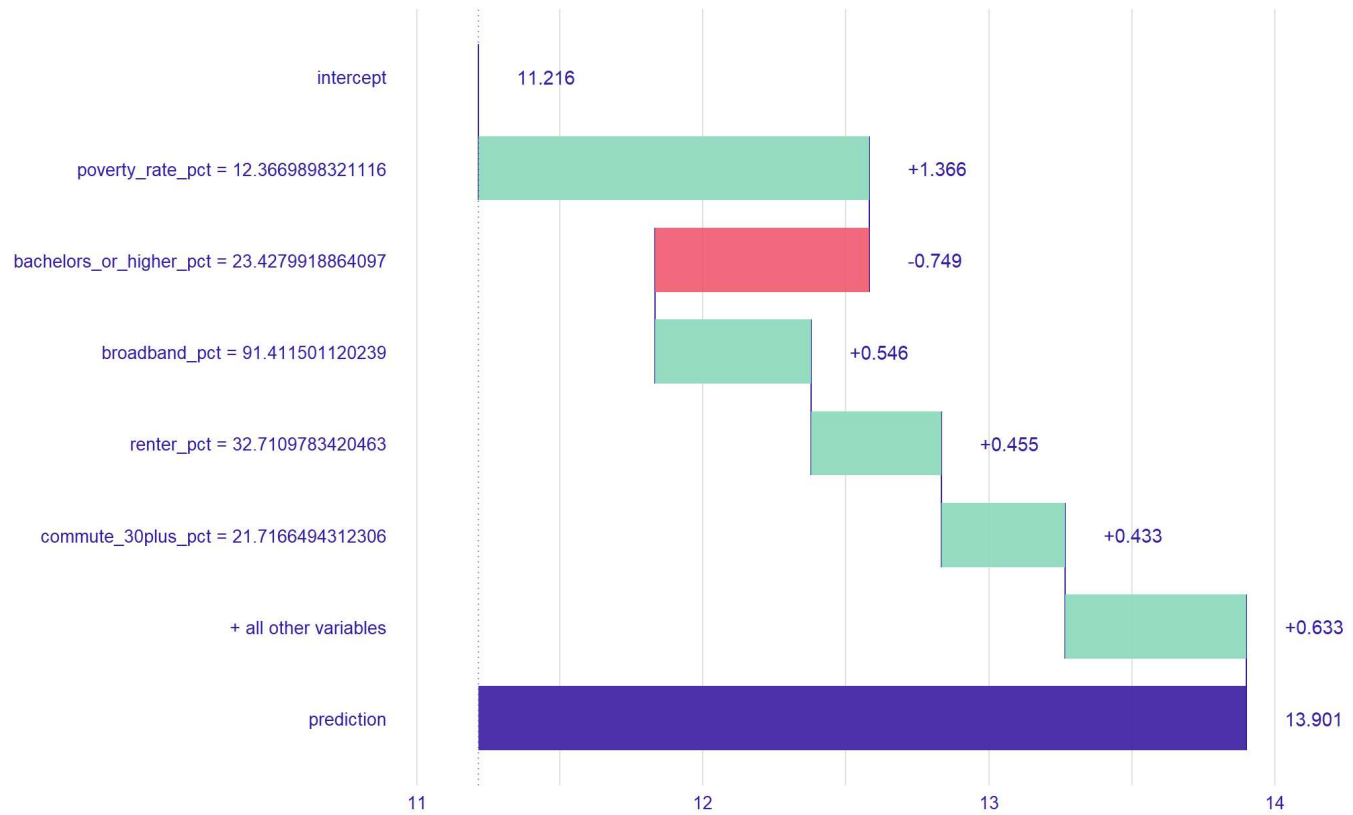
Examine what drives predictions for **Census Tract 304.04 on Maui, Hawai'i**.

Hints & Resources: - This tract is observation #154 in our data (hint: `obs = 154`) - Use: `plot_contribution()` - **Red bars** = pushing toward higher SNAP % - **Blue bars** = pushing toward lower SNAP % - **Bar length** = strength of effect

Your Code:

```
# Plot contribution effects for Census Tract 304.04 (Maui, observation #154)
plot_contribution(treesnap1, obs = 154)
```


SHAP Break-Down



Case Study Analysis

Reflection & Analysis

Final Reflection Questions

Answer these questions in text below:

Understanding the Analysis

- Model Performance:** How well does your Random Forest predict SNAP participation? What might explain the remaining unexplained variation? The model predicted SNAP participation fairly well. It captured most of the major patterns (~59%) but not everything. The remaining variation could be due to factors not included in the dataset.
- Key Drivers:** Based on feature importance and dependence plots, which 2-3 factors are most strongly associated with SNAP participation in Hawai‘i? The most important predictors of SNAP participation were poverty rate, median income, and unemployment rate. These make sense because lower income and higher unemployment usually lead to more people needing food assistance.
- Equity Implications:**
 - What does the geographic distribution of SNAP participation tell you about food security disparities?

- Are certain communities more vulnerable? The map showed that SNAP participation is higher in both rural and urban areas under economic pressure. This suggests that food insecurity in Hawai'i is shaped by both poverty and geography.

4. Limitations:

- What important variables might be missing from this analysis?
- What are the limitations of predicting SNAP participation? (Remember: correlation $\neq$ causation) This analysis doesn't include every factor that affects food insecurity, such as food prices, local store access, or community support programs. Also, the model only shows correlations, not direct causes.

5. Policy Recommendations:

- Based on your findings, what 2-3 policy interventions might help address food insecurity? Improving job access, supporting local food systems, and expanding outreach in high need areas.
- Who should be targeted? Where should resources be focused? Efforts should target both low income urban neighborhoods and remote rural communities where access and opportunity are limited.

Technical Reflection

6. **Machine Learning Interpretation:** How did SHAP values help you understand the Random Forest model compared to just looking at variable importance? They helped me see why the model made certain predictions. Instead of just knowing which variables were important, it showed me how each feature pushed the prediction higher or lower.
7. **Next Steps:** If you were to continue this analysis, what would you do next? I'd gather more detailed community data, like information on local food programs, transportation access, and housing costs.

★ BONUS: Extension Challenges

For Extra Credit

Try one or more of these extensions:

Extension 1: Regional Analysis

- Split Hawai'i by island (using **GEOID** or geographic location)
- Build separate Random Forest models for each island
- Compare which factors matter most on each island

Extension 2: Relationship Exploration

- Create scatter plots showing relationships between top predictors and SNAP%
- Calculate correlation coefficients
- Which relationships are strongest?

Extension 3: Model Comparison

- Build a **second** Random Forest with different parameters (e.g., **mtry**, **nodesize**)
- Compare performance to your original model

- Which performs better?

Extension 4: Prediction Intervals

- For a new hypothetical census tract, use your model to predict SNAP participation
- What would be needed to create prediction intervals?
- How confident would you be in that prediction?

Extension 5: Data Visualization

- Create a multi-panel visualization combining:
 - The map of SNAP participation
 - Distribution histogram
 - Top feature importance plot
 - Tell a complete visual story



Submission Checklist

- ☐ All 8 challenges completed with code
- ☐ Written explanations for each challenge
- ☐ Reflection questions answered thoroughly
- ☐ Code is commented and readable
- ☐ Plots are properly labeled with titles/legends
- ☐ At least one bonus challenge attempted (optional)
- ☐ **Rendered to PDF and submitted to Canvas**



How to Render & Submit

Step 1: Render to HTML

In RStudio, click the blue **“Render”** button at the top of the editor pane.

Wait for the rendering to complete - you'll see the document appear in the **Viewer pane** (bottom right).

Tip

Rendering typically takes 2-5 minutes depending on how fast your computer is. Be patient!

Step 2: Open in New Window

In the **Viewer pane**, click the **“Show in new window”** icon (it looks like an arrow pointing to a box, usually in the top-right corner of the viewer).

Your rendered HTML document will open in your default web browser.

Step 3: Print to PDF

1. In your browser, press **Ctrl + P** (Windows) or **Cmd + P** (Mac)
2. A print dialog will appear
3. Change the destination to **“Save as PDF”**
4. Give it a clear name: `LastName_FirstName_ml_assignment.pdf`
5. Click **“Save”**

Your PDF will download to your Downloads folder (or wherever your browser saves files).

Step 4: Upload to Canvas

1. Go to your Canvas course page
2. Find the assignment: “Hawaii SNAP Analysis Homework”
3. Click **“Submit Assignment”**
4. Click **“Choose File”** and select your PDF
5. Click **“Submit”**

Important Reminders

-  Make sure your `.qmd` file renders **without errors** before saving as PDF
-  Double-check that all your challenge code is visible in the HTML output
-  Your PDF should include both your code AND the results/plots
-  Save the PDF file somewhere you can find it (Desktop or a course folder)
-  Allow 5-10 minutes total for rendering and saving

Troubleshooting

“Render button is grayed out?” - Make sure you have the `.qmd` file open in the editor (not just viewed)

“I don’t see a Viewer pane?” - Go to **View** → **Panes** → **Show Viewer Pane**

“Nothing appears after clicking Render?” - Check the R console for error messages - Make sure all your code runs without errors first - Try re-rendering after fixing any issues



Resources & References

- [tidycensus Documentation](#)
- [Census Bureau Data Catalog](#)
- [SHAP Documentation](#)
- [Random Forests in R](#)
- [mapview Package](#)

Good luck! Remember: focus on understanding WHAT you’re doing and WHY, not just running code. Ask questions, explore the data, and enjoy the detective work of data science!  

